

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 1

Course Code: INT396

Course Title: UNSUPERVISED LEARNING

Course Outcome: CO1

Paper: Sample Subjective Questions

Maximum Marks: 10 x 10 = 100

Unit: 1

Unit Title: Foundations of Unsupervised Learning

Unit Mapped RBT: L2 (Understand)

Total Questions: 10

Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Explain why unsupervised learning is used at all when a supervised method would give a more precise answer. (10 Marks)

- (a) Describe two situations in which the labels a supervised method needs cannot be obtained.
- (b) Explain what an unsupervised method is still able to tell the analyst in those situations.

[L2 - Understand]

Q2. Describe the part that the distance measure plays in an unsupervised method. (10 Marks)

- (a) Describe how the choice of measure changes which records are treated as alike.
- (b) Explain why two analysts running the same algorithm on the same table can report different groups.

[L2 - Understand]

Q3. Distinguish Euclidean distance from cosine similarity for the comparison of customer purchase records. (10 Marks)

- (a) Describe what each of the two actually responds to in a pair of records.
- (b) Explain which of the two you would use to compare spending patterns while ignoring the overall amount spent, and give the reason.

[L2 - Understand]

Q4. Explain how market segmentation is carried out on an unlabelled customer table. (10 Marks)

- (a) Describe the features you would use and the preparation those features need.
- (b) Explain what a business actually does with the groups once they have been produced.

[L2 - Understand]

Q5. Describe anomaly and fraud detection as an unsupervised problem. (10 Marks)

- (a) Describe what makes a record anomalous when no fraud labels are available.
- (b) Explain why a very small number of frauds among very many transactions makes the task difficult.

[L2 - Understand]

Q6. Interpret the following situation. A retailer clusters its customers using age, income and postcode, with the postcode entered as a plain number, and the groups turn out to follow the postcode alone. (10 Marks)

- (a) Describe what has gone wrong and why the postcode has taken over the result.
- (b) Explain the two corrections that are needed before the clustering is repeated.

[L2 - Understand]

Q7. Predict what happens to a clustering when features measured on very different scales are used without any preparation. (10 Marks)

- (a) Describe the effect on the distances between records.
- (b) Explain how the problem is corrected, and what changes in the result once it is.

[L2 - Understand]

Q8. Summarize how pattern discovery is applied to biological data and to social data. (10 Marks)

- (a) Describe what the records and the features are in each of the two cases.
- (b) Explain the kind of finding the analyst is hoping for in each of the two.

[L2 - Understand]

Q9. Explain what an analyst should report about a clustering when no accuracy figure can be produced. (10 Marks)

- (a) Describe two things that can be reported in place of accuracy.
- (b) Explain what each of the two tells the reader about the quality of the result.

[L2 - Understand]

Q10. Recognize what has to be settled before an unsupervised project begins. (10 Marks)

- (a) Describe the decisions that must be made about the features and about similarity.
- (b) Explain the consequence of choosing the features badly, and why a better algorithm cannot repair it.

[L2 - Understand]

----- End of Sample Question Set -----